

HO�F–RIESZ: Round 004

A diffusion-retaining local pair inverse

Campaign research memorandum

17 September 2026

Published preceding checkpoint: 52bda5d0d24067b051c6fe9763f2a78e7599e593. The construction and independent reconstruction use separately sealed inputs and reports. Their precise audit dispositions are recorded in the round report. The result below concerns a local auxiliary pair process and an initial iid diagnostic. Classical regularity and the full singular fluctuation theorem are not asserted.

1 The radial comparison

Fix an integer $d \geq 3$, $0 < s \leq d - 2$, $p = s + 2$, $N \geq 2$, $0 \leq \nu < \infty$, $0 \leq T < \infty$, and a real symmetric matrix A . On $D = \mathbb{R}^d \setminus \{0\}$, put

$$v_N(z) = \frac{2s}{N}|z|^{-s-2}z, \quad L = 2\nu\Delta + v_N \cdot \nabla, \quad j_0(z) = s|z|^{-s}, \quad J_A(z) = j_0(z)\theta^T A \theta, \quad \theta = z/|z|.$$

The terminal equation is $(\partial_t + L)\Phi = -J_A$, with $\Phi(T, \cdot) = 0$. The coefficient 2ν is the relative diffusion, corresponding to noise $\sqrt{4\nu}$. Define

$$F_{N,\tau}(r) = \frac{N}{4} \left[(r^p + 2sp\tau/N)^{2/p} - r^2 \right], \quad r > 0.$$

For $q = r^p/(r^p + 2sp\tau/N)$, direct differentiation gives

$$\partial_\tau F - v_N \cdot \nabla F = sr^{-s}, \tag{1}$$

$$\Delta F = \frac{N}{2} \left[q^{s/p}(d + s - sq) - d \right]. \tag{2}$$

The derivative of $q^{s/p}(d + s - sq)$ is

$$\frac{s}{p}q^{s/p-1}[d + s - (2s + 2)q] \geq 0,$$

since $d \geq s + 2$. Its value at $q = 1$ is d , so $-Nd/2 \leq \Delta F \leq 0$, strictly negative for positive remaining time. Consequently

$$(\partial_t + L)F_{N,T-t} = -j_0 + 2\nu\Delta F \leq -j_0.$$

Writing $K = \|A\|_{\text{op}}$, this is a radial majorant for the signed source, because $|J_A| \leq Kj_0$. The radial amplitude satisfies

$$0 \leq F_{N,\tau}(r) \leq \min \left\{ \frac{(2sp)^{2/p}}{4} N^{s/p} \tau^{2/p}, \quad s\tau r^{-s} \right\}. \tag{3}$$

No small-diffusion assumption enters this inequality. The anisotropic transport solution itself need not be superharmonic.

2 Construction, collision avoidance and source integrability

Construct the path solving

$$Z_h = z + \sqrt{4\nu}B_h + \int_0^h v_N(Z_a) da$$

by bounded Lipschitz cutoffs of the drift, stopped on increasing annuli. For each cutoff, Picard increments on $[0, H]$ are bounded by $B_m L_m^k H^{k+1}/(k+1)!$. Their uniform convergence gives existence; the same factorial estimate gives uniqueness. The resulting solution map is continuous in its initial point and continuous driving path for the cutoff equation. Countable patching gives a Borel maximal map, with lifetime the limit of the annular exit times.

The following Itô calculations stop inside a finite annulus. All derivatives and coefficients of the stochastic integrals are bounded there. For $V(r) = r^2 + r^{-(d-2)}$,

$$LV = 4\nu d + \frac{4s}{N}r^{-s} - \frac{2s(d-2)}{N}r^{-s-d} \leq C_{N,\nu},$$

where an explicit admissible constant is

$$C_{N,\nu} = 4\nu d + \frac{4sd}{N(s+d)} \left[\frac{2s}{(d-2)(s+d)} \right]^{s/d}.$$

If $\rho_{a,R}$ is the first exit from $a < |z| < R$, expectation of stopped Itô's formula gives

$$\mathbb{P}_z(\rho_{a,R} \leq H) \leq (a^{d-2} + R^{-2})[V(z) + C_{N,\nu}H].$$

Taking $a \downarrow 0$, $R \uparrow \infty$ in nested annuli proves noncollision and nonexplosion from each nonzero starting point. It also identifies the patched global path, rather than merely extracting a subsequential law. The constant here need not be uniform in ν .

For a fixed remaining time τ , stopped Itô applied to $F_{N,\tau-h}(|Z_h|)$ gives

$$\mathbb{E}_z F_{N,\tau-\tau \wedge \rho}(|Z_{\tau \wedge \rho}|) + \mathbb{E}_z \int_0^{\tau \wedge \rho} j_0(Z_h) dh \leq F_{N,\tau}(|z|).$$

The first term is nonnegative. Monotone convergence of the positive source integrals, after noncollision has been established, proves

$$\mathbb{E}_z \int_0^\tau j_0(Z_h) dh \leq F_{N,\tau}(|z|). \quad (4)$$

This supplies expectation integrability, which would not follow from pathwise avoidance alone. The independent reconstruction also obtains the sharper separate exit bounds

$$\begin{aligned} \mathbb{P}_z(\rho_{a,R} \leq H, |Z_\rho| = a) &\leq (a/|z|)^{d-2}, \\ \mathbb{P}_z(\rho_{a,R} \leq H, |Z_\rho| = R) &\leq \frac{|z|^2 + 4\nu dH + (4/N)F_{N,H}(|z|)}{R^2}. \end{aligned}$$

They follow by differentiating r^{2-d} and r^2 , respectively.

3 The precise solution class and domain passage

The absolutely integrable potential

$$U_A(\tau, z) = \mathbb{E}_z \int_0^\tau J_A(Z_h) dh, \quad \Phi_A(t, z) = U_A(T - t, z)$$

is jointly Borel: time integration of the measurable path map followed by integration over Brownian paths preserves measurability; positive and negative parts are finite by (4). Moreover,

$$|U_A(\tau, z)| \leq K F_{N,\tau}(|z|). \quad (5)$$

This is a global bounded function for fixed N, T ; its supremum need not be uniform in N .

Pathwise uniqueness gives the deterministic flow identity under time shifts. Brownian independent increments then give the deterministic-time Markov property. Measurability permits conditioning on the random restart point: the exceptional future path set has zero probability for each fixed starting point, so its conditional integral is zero. Truncation and (4) extend this conditioning to the source. For each starting (t, z) ,

$$\Phi_A(t + h, Z_h) + \int_0^h J_A(Z_a) da = \mathbb{E}_z \left[\int_0^{T-t} J_A(Z_a) da \middle| \mathcal{F}_h \right].$$

Thus the left side is a true, uniformly integrable martingale. The claimed solution class consists exactly of bounded Borel functions with zero terminal value and this true-martingale property from every starting point. Taking terminal expectations proves uniqueness in this class. No differentiability or origin trace is part of its definition.

For $\nu > 0$, killing at annular exit gives the unique bounded killed martingale potential

$$U_A^{a,R}(\tau, z) = \mathbb{E}_z \int_0^{\tau \wedge \rho_{a,R}} J_A(Z_h) dh.$$

It obeys the same radial bound. On nested annuli the signed source integrals converge almost surely and in L^1 , dominated by the full integrable absolute source integral. Their expectations therefore converge to U_A at each fixed time and point. This proves the specified domain passage. Classical interior regularity is not inferred. Any bounded classical solution meeting the stopped-Itô and terminal-continuity hypotheses is identified with this potential by the same L^1 passage.

When $\nu = 0$, the path is the outward characteristic $(|z|^p + 2sph/N)^{1/p}\theta$. The potential equals $(\theta^T A \theta) F_{N,\tau}(|z|)$ and is unique in the pointwise forward absolutely-continuous characteristic class. No extra annular outflow condition is imposed. For $A = I$, $\nu > 0$ and $\tau > 0$, dominated convergence in the stopped equality gives the strict diagnostic

$$0 < U_I(\tau, z) = F_{N,\tau}(|z|) + 2\nu \mathbb{E}_z \int_0^\tau \Delta F_{N,\tau-h}(|Z_h|) dh < F_{N,\tau}(|z|).$$

Here (2) bounds the Laplacian in absolute value, so this passage needs no uncontrolled derivative limit.

4 The initial iid diagnostic

Fix a smooth even cutoff $0 \leq \chi \leq 1$, equal to one near zero and supported in an embedded torus ball B_R , $R < 1/2$. Set $H(x, y) = \chi(x - y)U_A(\tau, x - y)$, with any finite diagonal value. Reflection

symmetry makes this a real symmetric Borel pair kernel. For iid labels of density μ with $\|\mu\|_\infty \leq M$,

$$\|H\|_{L^2(\mu^2)}^2 \leq M \|\chi U_A\|_{L^2(\mathbb{T}^d)}^2 \leq MK^2 \int_{B_R} F_{N,\tau}^2.$$

There is one density factor because $\int \mu(y+z)\mu(y)dy \leq M$. For $m = \mu^2(H)$, $q = H_\mu - m$, and $H_c = H - m - q(x) - q(y)$, the exact mean-field-centered statistic is

$$P_N[H] = -\frac{m}{2N} - \frac{1}{N^2} \sum_i q(X_i) + \frac{1}{N^2} \sum_{i<j} H_c(X_i, X_j).$$

Conditional centering and shared-label orthogonality give

$$\mathbb{E}P_N[H]^2 = \frac{m^2}{4N^2} + \frac{\|q\|_2^2}{N^3} + \frac{N-1}{2N^3} \|H_c\|_2^2 \leq \frac{N-1}{2N^3} \|H\|_2^2.$$

This retains the deterministic bias and first projection. Put $b_N = \min(\beta_N, 1)$ and $\sigma_N = \sqrt{Nb_N}$. For $\ell = (2sp\tau/N)^{1/p}$, (3) yields

$$\int_{B_R} F^2 \leq |S^{d-1}| \left[\frac{N^2 \ell^4 \min(R, \ell)^d}{16d} + \mathbf{1}_{\ell < R} s^2 \tau^2 \int_\ell^R r^{d-1-2s} dr \right].$$

Consequently, for all $N \geq 2$ and $\tau \in [0, T]$,

$$\sup_{0 \leq \tau \leq T} \mathbb{E}|\sigma_N P_N[H]|^2 \leq C_T b_N \begin{cases} N^{-1}, & 2s < d, \\ (1 + \log N)/N, & 2s = d, \\ N^{-(d+2-s)/(s+2)}, & 2s > d. \end{cases}$$

The constant is independent of finite $\nu \geq 0$, $\beta_N > 0$ and N ; $\nu = 1/\beta_N$ is permitted. In the middle case $\tau^2 \log_+(1/\tau)$ is bounded at zero. A core larger than the cutoff ball is controlled by the first term of the preceding exact bound. The time supremum is outside the expectation. No lower bound for the diffusive norm or path supremum is claimed.

5 Remaining mathematical obligations

The constructed diagnostic does not solve the full periodic pair equation. Ordinary transport, both nonlocal responses, the actual time-dependent backward test, the periodic force remainder and cutoff equation errors still require a proof. Initial iid estimates do not transfer to the evolved interacting law. The cubic residual, contractions, pair bracket, singular generator passage and critical hierarchy tail remain open. The full singularity range $0 < s < d$ also extends beyond $s \leq d-2$ here.

Throughout, the old condition $\beta_N N^{2s/d-1} \rightarrow 0$ remains distinct from full microscopic subcriticality $\lambda_N = \beta_N N^{s/d-1} \rightarrow 0$ and from microscopic criticality $\lambda_N \rightarrow \lambda \in (0, \infty)$. This local estimate changes none of those normalizations. The logarithmic case requires its separate convention. The complete Markdown proofs, their immutable dossiers and the round report supply the detailed constants and actual audit status.